Primary Quadratic Modules for Heisenberg Fixed Fibres

Route-A structural certificate C156

Abstract

For a frozen Heisenberg lattice automorphism we determine the Smith type of every horizontal fixed-class group, reduce the universal central-rotation denominator to its group exponent, and split the resulting quadratic function into orthogonal primary components. Exact component ledgers through iterate fourteen turn the clean fixed-circle count into a product of primary zero counts. Primary is finite-group terminology here, not an arithmetic factorization.

Keywords: Heisenberg nilmanifold; finite quadratic module; Smith normal form; primary decomposition; fixed fibres; exact verification.

1 Smith form and the sharp universal bound

Let $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, $B = A^n$, and $M = B - I$. With F_n, L_n denoting Fibonacci and Lucas numbers, the standard doubling and Cassini identities give

$$M = \begin{cases} L_n \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}, & n \text{ odd}, \\ F_n \begin{pmatrix} L_{n+1} & L_n \\ L_n & L_{n-1} \end{pmatrix}, & n \text{ even}. \end{cases} \tag{1}$$

The cofactors have determinants -1 and -5, respectively; the latter has content one for even n. Consequently

$$G_n = \mathbb{Z}^2/M\mathbb{Z}^2 \cong \begin{cases} (\mathbb{Z}/L_n\mathbb{Z})^2, & n \text{ odd}, \\ \mathbb{Z}/F_n\mathbb{Z} \times \mathbb{Z}/(5F_n)\mathbb{Z}, & n \text{ even}. \end{cases} \tag{2}$$

Its exponent is $h_n = L_n$ for odd n and $h_n = 5F_n$ for even n.

For $B = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, the canonical lattice correction is

$$q_B(x, y) = \frac{ac}{2} x(x-1) + bcxy + \frac{bd}{2} y(y-1). \tag{3}$$

The actual iterate correction satisfies $q_n = q_{A^n} + \ell_n$, where ℓ_n is integer linear. For $m = Mv$, define $\rho_n([m]) = q_n(v) - m_1 v_2 \pmod 1$. Writing $M = gU$ and scaling v by the exponent, direct substitution in (3), followed by the period-three parity of F_n, L_n modulo two, proves

$$h_n \rho_n([m]) = 0 \quad \text{in } \mathbb{Q}/\mathbb{Z}. \tag{4}$$

This is an all-iterate divisibility result. Equality of the observed denominator with h_n is recorded only for $2 \le n \le 14$.

2 Orthogonal primary decomposition

The polarization of ρ_n is bilinear. Hence primary components of G_n at distinct primes are orthogonal, and the restriction to a component of exponent p^e takes values in $p^{-e}\mathbb{Z}/\mathbb{Z}$. Uniqueness of primary torsion in $\mathbb{Q}/\mathbb{Z}$ gives

$$C_n = \prod_{p \mid h_n} C_{n,p}, \qquad C_{n,p} = \frac{1}{p^e} \sum_{a \bmod p^e} \sum_{x \in G_{n,p}} e^{2\pi i a \rho_n(x)}. \tag{5}$$

The formula is a finite root-of-unity zero projector, not an operator trace formula or an arithmetic local factor.

3 Exact certificate and boundary

CRT idempotents applied to the two standard generators, followed by column Hermite reduction, enumerate each primary component exactly. The counts are

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14
C_n	1	1	4	1	21	4	57	1	148	105	397	144	1041	57

The producer, direct-cocycle checker, and SymPy path independently verify the ledger. The strict tuple is (`A1_FAIL`, `A2_FAIL`, `A3_FAIL`, `A4_FORMAL_HINT`). Fixed sets remain clean circles and the ordinary isolated stability factor remains singular. We claim no isolated primitive-orbit determinant, target divisor, target functional equation or counting law, arithmetic local/Euler factor, root number, automorphy, Hilbert–Pólya construction, or Route-B authorization. Scope: `NO_BAD_EULER_OR_ROOT_NUMBER`.